

ONLINE SHOPPING COMPARISON BOT

A PROJECT REPORT

Submitted by

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ABSTRACT

“Online Shopping Comparison Bot” project utilizes UiPath Robotic Process Automation (RPA) to streamline product searches on Amazon by automating the retrieval of key details such as the product name, price, and ratings. Designed for efficiency, the workflow begins by terminating existing browser sessions to ensure a clean slate, followed by launching the Amazon website in Microsoft Edge. Upon receiving a product name as input, it enters the query in the search bar, executes the search, and extracts data from the first result using activities like Type Into, Click, and Get Value. Dynamic selectors enable precise interaction with web elements, while variables and arguments facilitate seamless input-output management. The workflow’s adaptability ensures it remains effective despite potential changes in Amazon’s webpage structure. With structured output, the solution is ideal for decision-making or integration into reporting systems. By automating repetitive e-commerce tasks, it saves time and effort, demonstrating the power of RPA in everyday applications. This project offers a scalable foundation for further enhancements like error handling, multi-product extraction, and integration with other platforms.

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LIST OF ABBREVIATIONS

ABBREVIATION	ACCRONYM
RPA	Robotic Process Automation
AI	Artificial Intelligence
API	Application Programming Interface
CV	Computer Vision
OCR	Optical Character Recognition

CHAPTER 1

INTRODUCTION

1.1INTRODUCTION

In today's digital era, online shopping has become a cornerstone of consumer behavior, with platforms like Amazon offering a vast array of products. However, manually searching for products, comparing their details, and extracting essential information can be time-consuming and repetitive. This challenge creates an opportunity for automation to enhance efficiency and accuracy, particularly for frequent or large-scale product searches. Leveraging technology to address this issue not only saves time but also ensures consistent and reliable data extraction.

Robotic Process Automation (RPA) has emerged as a transformative solution for automating repetitive tasks, making it an ideal approach for automating Amazon product searches. Using UiPath, a leading RPA tool, this project automates the process of searching for a product on Amazon and retrieving key details such as the product name, price, and ratings. By eliminating the need for manual intervention, this solution reduces human error and streamlines the workflow, empowering users to focus on decision-making rather than data gathering.

The workflow is designed to handle user input dynamically and interact with the Amazon webpage using selectors and activities like Type Into, Click, and Get Value. It begins by clearing any existing browser sessions to ensure a clean start, opens the Amazon website, and searches for the desired product based on user-provided input. Data extraction is performed efficiently and outputted in a structured format, ready for analysis or integration into other systems. The project showcases the ability of RPA to adapt to dynamic environments while maintaining reliability.

This project not only highlights the practical application of UiPath RPA in automating routine tasks but also demonstrates its potential to scale for more complex use cases. With a user-friendly design and robust implementation, it can be extended to handle bulk queries, integrate advanced error-handling mechanisms, or even combine with AI-based systems for enhanced functionality. By addressing a common challenge in e-commerce, the project underscores the growing significance of automation in simplifying workflows and maximizing productivity.

1.2 OBJECTIVE

The objective of this project is to develop an automated workflow for extracting product details from e-commerce platforms like Amazon, including product name, price, ratings, and images. It aims to eliminate manual search efforts and enhance efficiency by leveraging UiPath for seamless data retrieval. This automation is designed to improve accuracy, reduce time consumption, and streamline the product comparison process. Additionally, it seeks to demonstrate the capabilities of RPA tools in handling repetitive web-based tasks. Ultimately, it provides a robust foundation for scalable, real-world e-commerce automation solutions.

3. EXISTING SYSTEM

The existing system for extracting product details from e-commerce platforms relies heavily on manual effort. Users search for products, browse through multiple pages, and record details like price, ratings, and specifications manually, which is time-consuming and error-prone. This process lacks efficiency, scalability, and consistency, especially for large-scale data retrieval. Additionally, manual operations are prone to inaccuracies due to human oversight. The current approach is unsuitable for handling repetitive tasks effectively in dynamic e-commerce environments.

4. PROPOSED SYSTEM

This project proposes an automation solution using UiPath to streamline the process of product searches and data extraction from Amazon. The workflow begins by ensuring no active browser instances are running, ensuring a clean environment for the automation. It then opens the Amazon website in Microsoft Edge, where it dynamically enters the search query into the search bar. UiPath activities such as Type Into, Click, and Get Value interact with web elements to retrieve key product details, including the product name, price, and rating. Selectors are used to locate and extract this data, allowing the system to adapt to any changes in Amazon's webpage layout. The process utilizes variables and arguments to

efficiently manage inputs and outputs, returning the extracted information in a structured format suitable for further use in reporting or decision-making systems. This automated solution eliminates the need for repetitive manual searches, improving efficiency and reducing time spent on data collection. Error handling mechanisms are in place to ensure the system runs smoothly, even if issues arise during execution. Additionally, dynamic selector management allows the workflow to remain functional despite changes in Amazon's layout. The data can be easily integrated into other tools for deeper analysis, such as business intelligence platforms. This approach also serves as a foundational model for automating similar e-commerce processes across other platforms. The workflow can be scaled to handle multiple queries simultaneously, offering batch processing capabilities. Overall, this project demonstrates the power of robotic process automation (RPA) in enhancing productivity and simplifying e-commerce tasks.

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CHAPTER 2

LITERATURE REVIEW

2.1 Survey on Robotic Process Automation (RPA) in Education:

Robotic Process Automation (RPA) is increasingly recognized as a valuable resource in education, streamlining teaching processes. For instance, RPA has been successfully implemented in grading assignments and managing student records, significantly reducing the workload of educators. However, challenges remain, particularly in automating tasks that require social interaction and adaptation to individual learning needs. The literature review of research papers related to RPA in Education is listed below:

The research discusses the rise of Artificial Intelligence (AI), robotics, and other digital technologies are creating a demand for new professions with evolved digital skills. Educational institutions must adopt these technologies to promote digital skills development and empower students to lead active and creative digital lives. The education sector is ready to witness a revolution with robotics process automation (RPA) technology. RPA focuses on the elimination of inefficiencies and the effort of human resources that is wasted while executing mundane tasks. RPA helps teachers, educators, students as well as parents directly or indirectly.

A research paper from IJITEE proposes a Robotic Process Automation (RPA) solution for the education domain. It shows the automation process for result analysis of student's examination results. The automation process takes input as the university result in pdf form. The research concludes that RPA

can help out here by saving time and under a budget which is a limited & crucial resource for educational institutes.

2.2 Survey on AI-Generated Content Detection:

AI-generated content detection has been a subject of substantial research. Existing tools such as OpenAI's GPT-3 and Google's BERT can discern between human-written and AI-generated text to a certain extent. However, these tools face challenges in accuracy, sometimes misclassifying AI-generated content as human-written. Techniques such as machine translation and content obfuscation further complicate the detection of AI-generated text. The literature review of research papers related to AI-Generated Content Detection is listed below:

The study investigates the capabilities of various AI content detection tools in discerning human and AI-authored content. Fifteen paragraphs each from ChatGPT Models 3.5 and 4 on the topic of cooling towers in the engineering process and five human-written control responses were generated for evaluation. The researchers conclude that the available detection tools are neither accurate nor reliable and have a main bias towards classifying the output as human-written rather than detecting AI-generated text.

A study conducted by the student of University of North Carolina seeks to evaluate the effectiveness of machine learning algorithms in differentiating between human-written and AI-generated text states that to accomplish this, responses from Computer Science students for both essay and programming assignments were collected. Several machine learning models, including Logistic Regression (LR), Decision Trees (DT), Support

Vector Machines (SVM), Neural Networks (NN), and Random Forests (RF), were trained and evaluated based on accuracy, computational efficiency, and confusion matrices.

2.3 Survey on Plagiarism Detection:

Plagiarism detection is a dynamic field with various tools and techniques. While these tools assist in identifying plagiarized content, they are not foolproof and may sometimes flag non-plagiarized material. For instance, detecting paraphrased content or content translated from another language remains a significant challenge. Continuous development and refinement of plagiarism detection tools are essential to address the evolving nature of plagiarism in academic submissions. The literature review of research papers related to Plagiarism Detection is listed below:

The article by [Norman Meuschke](#) summarizes the research on computational methods to detect academic plagiarism by systematically reviewing 239 research papers published between 2013 and 2018. The research covers 12 publicly available tools and two commercial systems (Turn it in and Plagiarism Check) that are widely used in the academic setting. The researchers conclude that the available detection tools are neither accurate nor reliable and have a main bias towards classifying the output as human-written rather than detecting AI-generated text.

The survey by Tezpur University presents a taxonomy of various plagiarism forms and includes discussion on each of these forms. Over the years, a good number of tools and techniques have been introduced to detect plagiarism. This paper highlights a few promising methods for plagiarism detection based on machine learning techniques.

2.4 Summary of the intersection of RPA, AI Detection, and Plagiarism

Checks:

“Online Shopping Comparison Bot ” bridges these areas, leveraging RPA to automate the product of e-commerce website. The project incorporates advanced algorithms for AI detection to identify potential AI-generated content. Additionally, it interfaces with external services for plagiarism checks, addressing the multifaceted challenges of maintaining academic integrity. This integration positions the project at the forefront of addressing contemporary concerns in the educational landscape. The project’s innovative approach aligns with the ongoing discourse in academia, presenting a timely solution to the challenges posed by AI-generated content and plagiarism. The amalgamation of RPA, AI detection, and plagiarism checks showcases the project’s significance in contributing to the academic integrity discourse.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM FLOW DIAGRAM

A flowchart is a type of diagram that represents an algorithm, workflow or process. The flowchart shows the steps as boxes of various kinds, and their order by connecting the boxes with arrows. This diagrammatic representation illustrates a solution model to a given problem. The system flow diagram for this project is in Fig. 3.1.

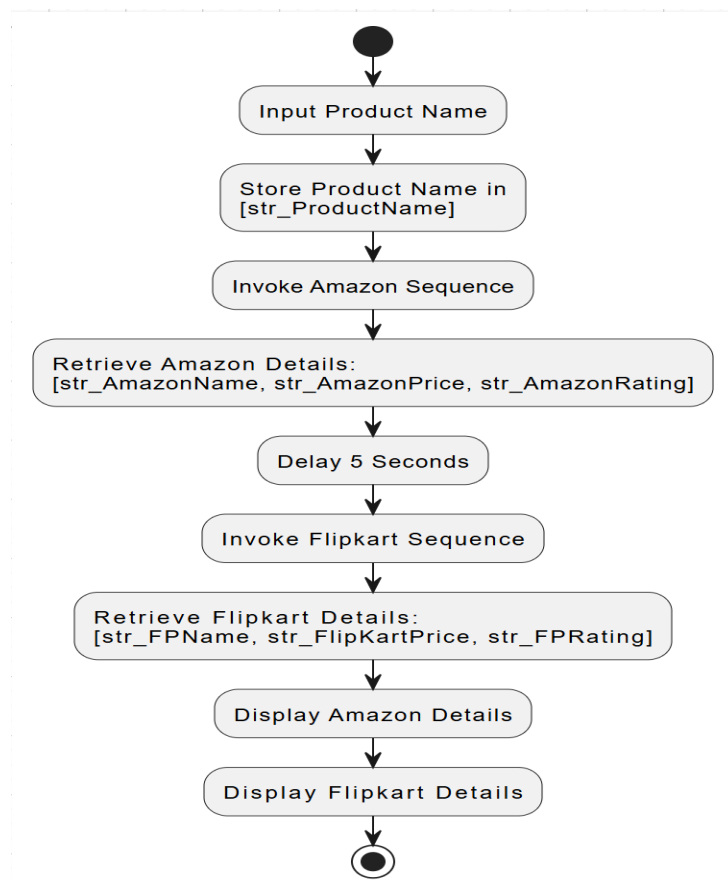


Fig 3.1 System Flow Diagram

3.2 ARCHITECTURE DIAGRAM

An architecture diagram is a graphical representation of a set of concepts, that are part of an architecture, including their principles, elements and components. The architecture diagram for this project is in Fig. 3.2.

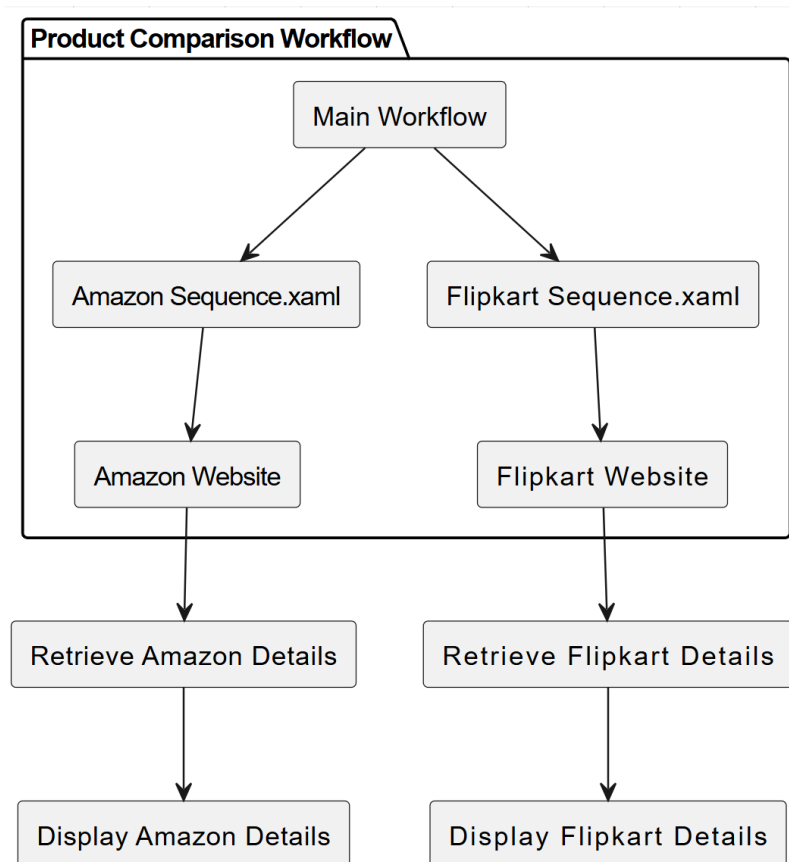


Fig 3.2 Architecture Diagram

3.3 SEQUENCE DIAGRAM

A sequence diagram is a type of interaction diagram because it describes how in what order a group of objects works together. This sequence diagram for this project is in Fig. 3.3.

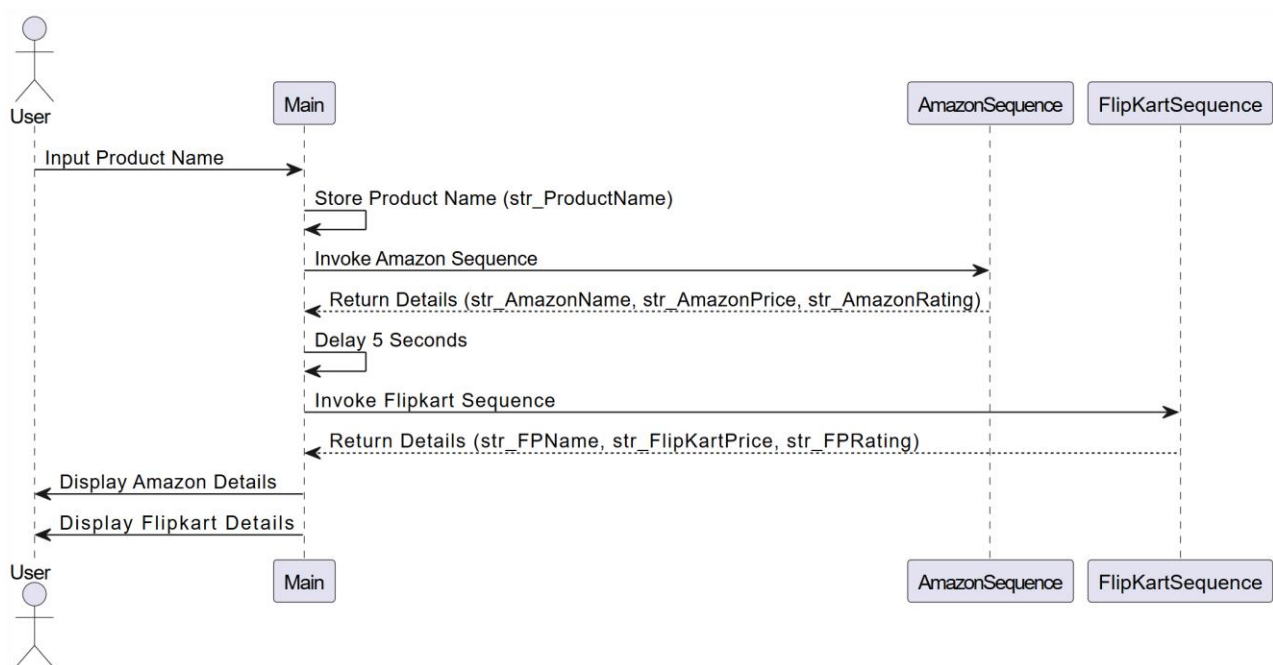


Fig 3.3 Sequence Diagram

CHAPTER 4

PROJECT DESCRIPTION

The "Online Shopping Comparison Bot" automates the process of comparing product details from popular e-commerce platforms like Amazon and Flipkart. It takes a product name as input and retrieves pricing, ratings, and names for the specified product from both platforms. The bot utilizes individual workflows for each website to scrape data and consolidates the results for display. Users can quickly compare essential product details to make informed purchasing decisions. This streamlines the shopping experience, saving time and effort.

1. MODULES:

1. INPUT HANDLING AND INITIALIZATION:

1. User Input Capture:

- The user is prompted via an input dialog to enter the product name.

2. Variable Initialization:

- Variables like AmazonPrice and FlipKartPrice are initialized to store product prices.
- Additional variables like AmazonRating and FlipKartRating are set up to hold product ratings.

3. Preparing for Workflow Execution:

- The initialized variables and user input are prepared for workflow invocation.

2. CONTENT ANALYSIS:

1. AI Detection:

- Iterate through each Word document in the selected subfolder.

- Employ advanced algorithms to detect AI-generated content.

2. Plagiarism Check:

- Interface with external plagiarism detection services.
- Conduct plagiarism checks on the assignment content.

3. RESULT MANAGEMENT:

1. Result Storage:

- The comparison results, stored in predefined variables for easy access and display.

2. Real-time Update:

- Display real-time updates of the integrity verification process.

4. COMPLETION AND REPORTING:

1. Completion Message:

- Conclude the operation with a message indicating the successful completion of the integrity verification task.

CHAPTER 5

OUTPUT SCREENSHOTS



Fig 5.1 – Input Dialog

The bot will get the product name in the input dialog box shown in Fig 5.1.

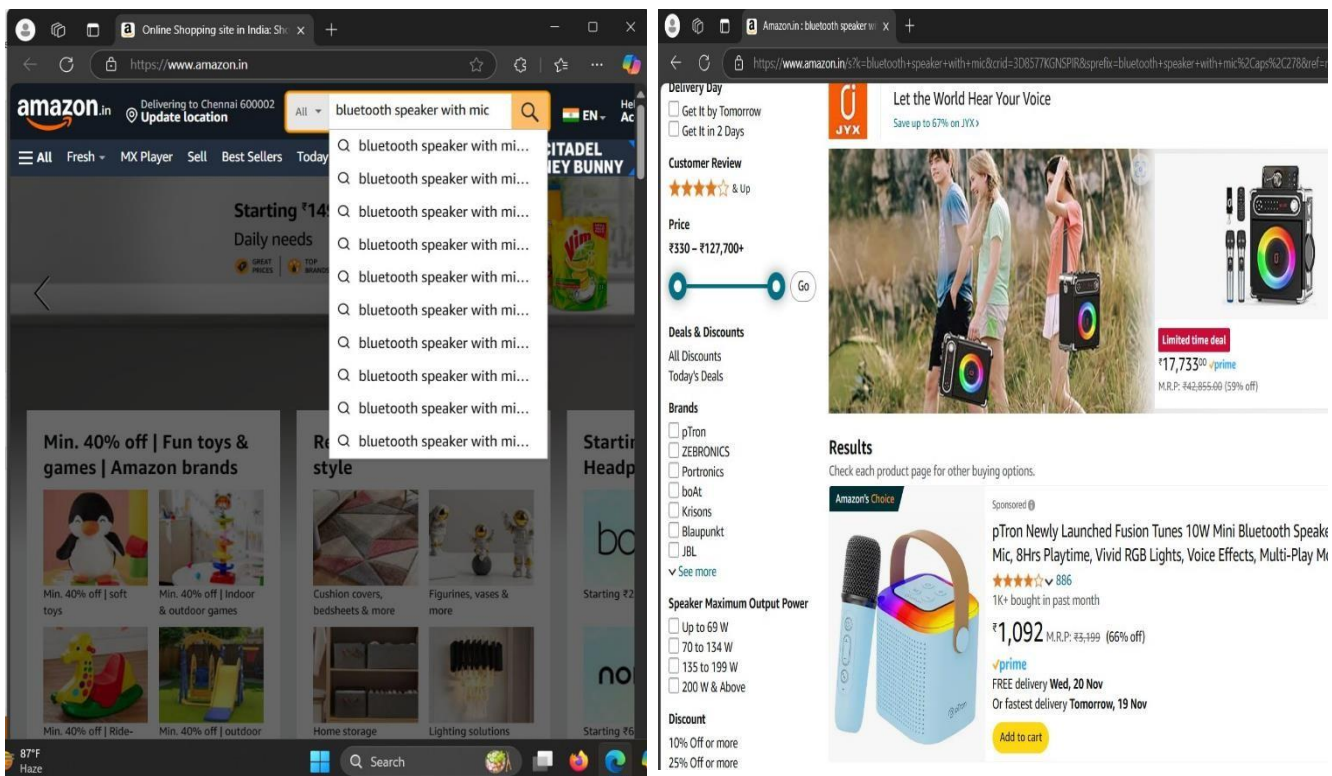


Fig 5.2 – Search for the product (Amazon)

The bot will search for the product in amazon and then fetch the data from the product as shown in the Fig 5.2.

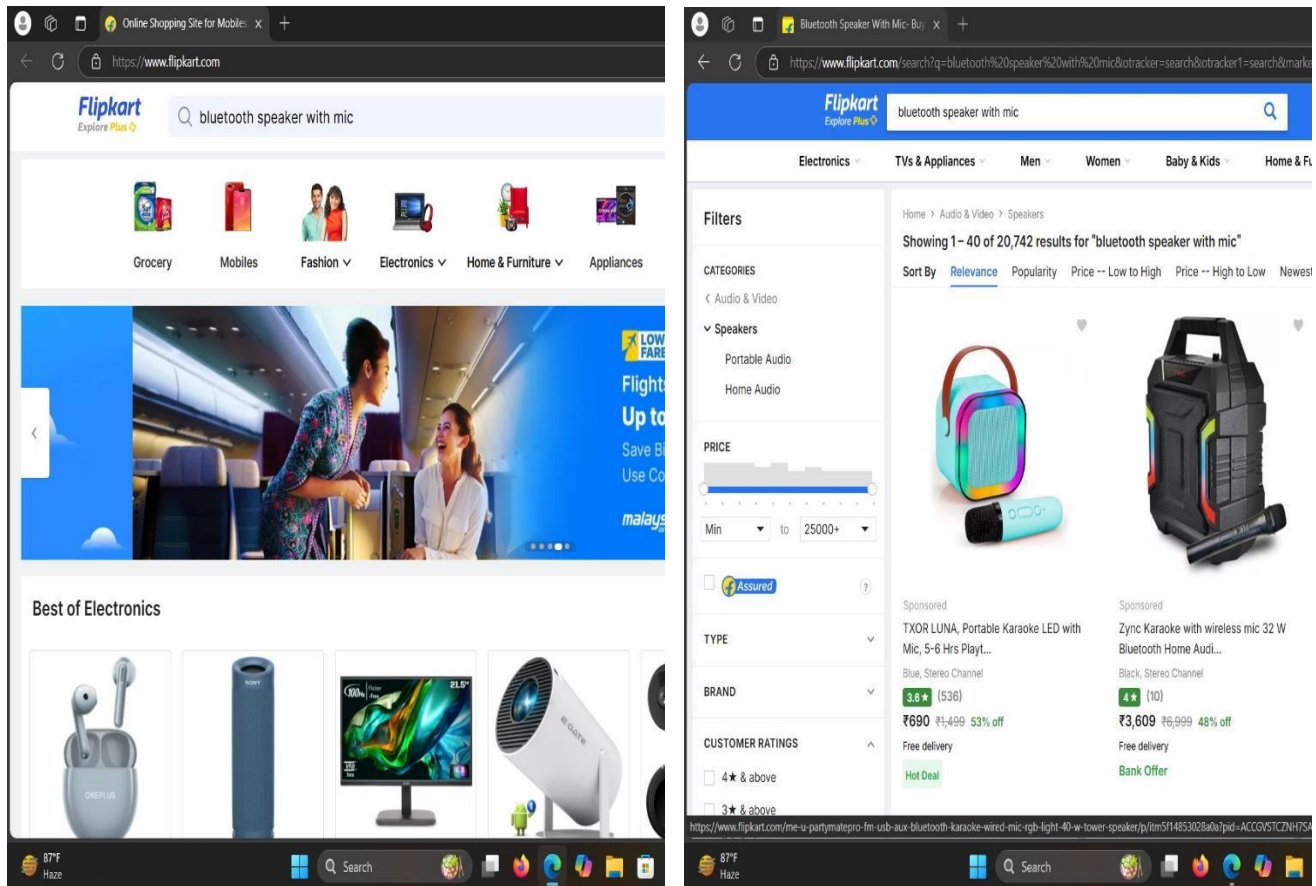


Fig 5.3 – Search for the product (flipkart)

The bot will search for the product in flipkart and then fetch the data from the product as shown in the fig 5.3

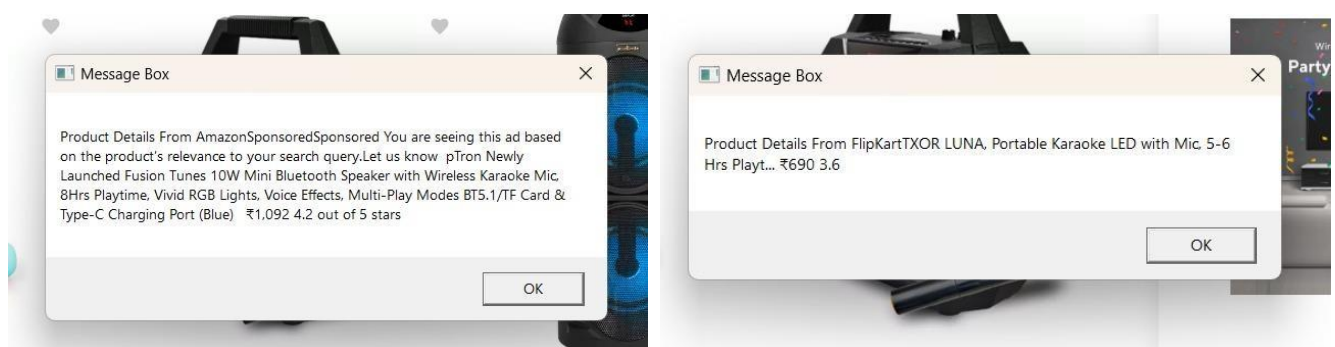


Fig 5.4 – Shows the Result in msg box

The bot will show the best product detail from flipkart and amazon website as shown in fig 5.4

CHAPTER 6

CONCLUSION

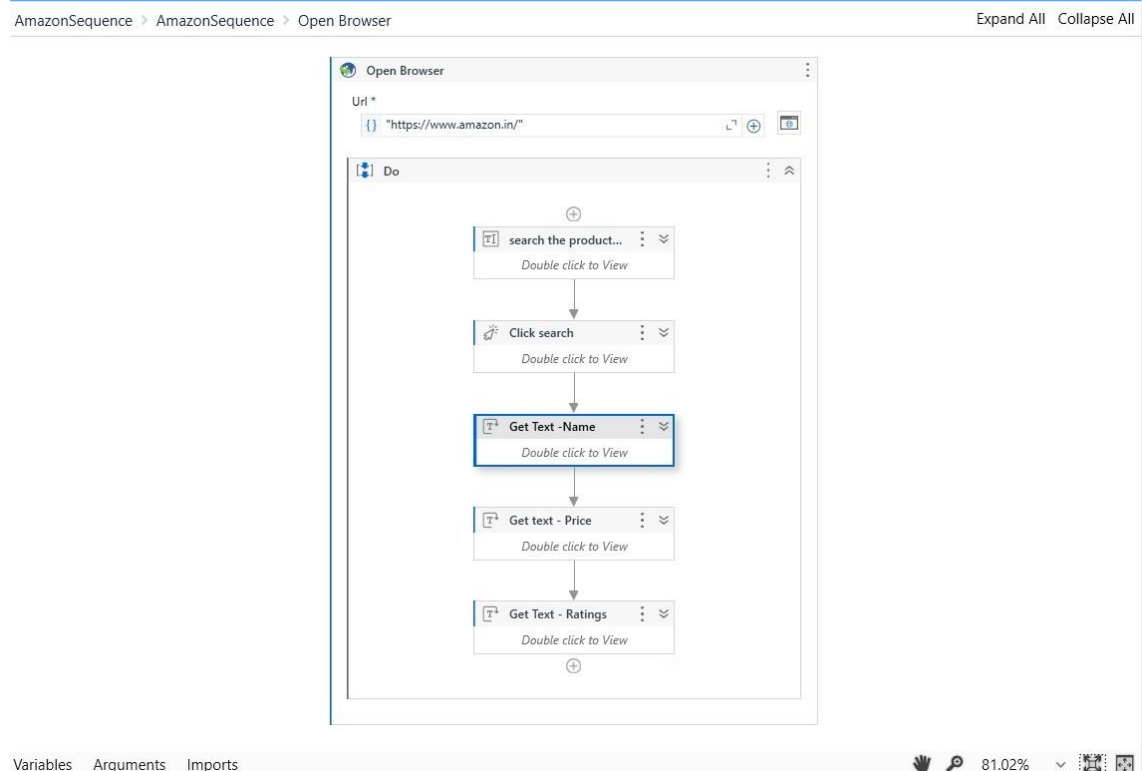
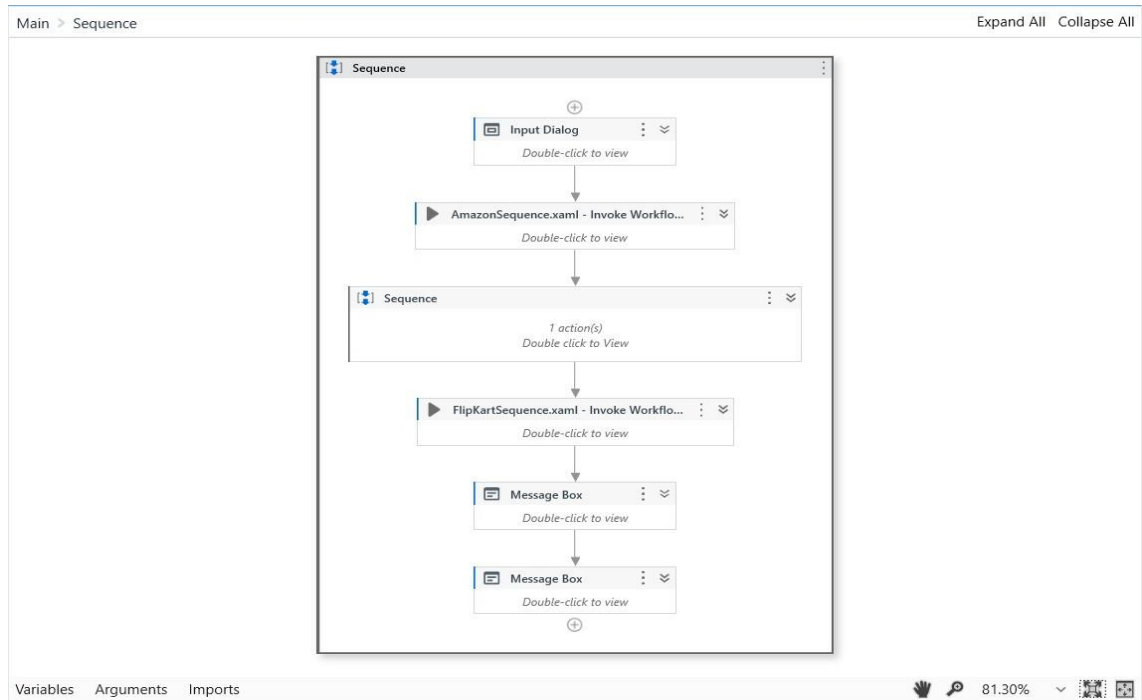
The “Online Shopping Comparison Bot” effectively simplifies the process of comparing products across major e-commerce platforms like Amazon and Flipkart. By automating data retrieval, it reduces the time and effort required for users to gather information about product prices, ratings, and names. The bot ensures accuracy and reliability by using robust workflows and precise data extraction techniques tailored for each platform. This automation enhances decision-making for users, offering a convenient solution for informed online shopping.

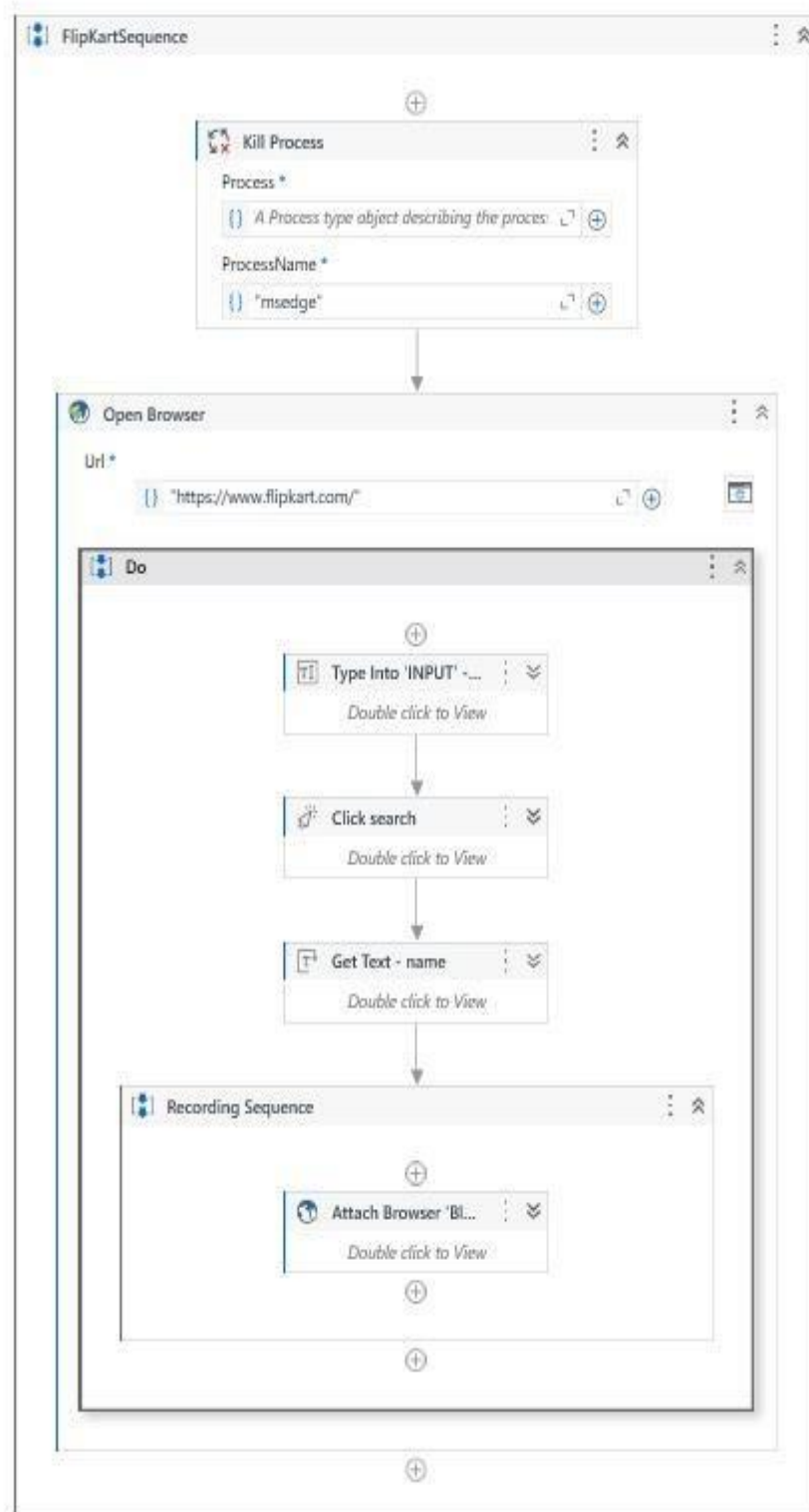
Moreover, the modular design of the bot facilitates easy updates and scalability, enabling it to adapt to future changes in e-commerce websites or expand its functionality. The segregation of workflows for Amazon and Flipkart ensures maintainability and modularity. Users benefit from a simple interface that hides complex backend operations, making the bot accessible to non-technical individuals. This approach ensures long-term usability and relevance in an ever-evolving e-commerce landscape.

Overall, the project demonstrates the power of automation in addressing real-world problems, highlighting the potential of tools like UiPath for intelligent workflows. The “Online Shopping Comparison Bot” not only improves the shopping experience but also sets a foundation for more advanced applications in data comparison and automation. It showcases the potential to evolve into a broader price comparison platform for additional markets, providing even greater value to consumers.

APPENDIX

PROCESS WORK FLOW





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